

Parametric Expansion of Clear-Sky Beam Solar Irradiance into Trigonometric Time-Series

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Abstract

- Direct Beam Solar Irradiance (*DBSI*), defined as a product of direct normal irradiance and cosine of solar zenith distance, is expanded into fast converging trigonometric series. The amplitudes and phases of individual trigonometric terms are expressed as functions of the Earth's orbital eccentricity, perihelion longitude and obliquity, while the frequencies are linear combinations of solar hour angle frequency and multiples of solar longitude frequency.
- The *DBSI* trigonometric series are continuous functions of time, globally applicable, i.e. valid for any geographic latitude and longitude with arbitrary accuracy. If truncated after the third term the maximum error varies seasonally in $\pm 0.4 \text{ W/m}^2$ interval.
- The *DBSI* trigonometric series can be readily used to calculate solar irradiance using time-varying orbital elements. Further, their application could be extended to calculations of irradiance on the surface of other celestial bodies.

Keywords: Direct Beam Solar Irradiance, Trigonometric Series, Orbital Elements, Atmospheric Transmittance.

1. Introduction and Background

With advent of photovoltaic and thermal applications of solar energy the search for globally applicable while arbitrarily accurate approximations of

the surface solar irradiance function is still ongoing. Generally, surface solar irradiance is affected by three major factors:

- Sun’s energy output (Sun’s luminosity)
- Earth’s orbital motion and orientation of its rotation axis
- Earth’s atmosphere

Sun’s luminosity varies only $\approx \pm 0.2\%$ over 11 year long cycle. The changes in Earth’s orientation due to its orbital motion cause annual variation of $\approx \pm 4\%$ in the incoming radiant energy. Finally, each day during daylight hours, between 25% (for low Sun’s zenith distances) to $> 85\%$ (for Sun close to the horizon), of incoming solar energy is altered or stopped by the earth’s atmosphere. The Earth’s orbit and orientation drive the seasonal character of irradiance, and therefore the physical properties of climate. Since solar irradiance as main driver of the climate, is most sensitive to changes in Earth’s atmosphere it is not surprising that modeling of the radiation transfer phenomena through the atmosphere and its impact on Earth’ climate, gets most of the attention in the scientific literature.

Despite the central role of *DBSI* in climatology, solar energy engineering, and planetary science, most existing formulations rely on empirical coefficients, location-specific tuning. These approaches often lack global generality, or introduce discontinuities at sunrise and sunset. As a result, there is a need for a compact, analytic representation of *DBSI* that remains valid for any latitude, any day of the year, and any planetary body with known orbital elements.

The main objective of this paper is to capture, in analytical form, dependence of the solar irradiance on the Earth’s orbital parameters, for any point on Earth. In accepted terminology [NREL 2023] solar irradiance received at the ground level - Global Horizontal Irradiance (*GHI*), consists of two components: Diffuse Horizontal Irradiance (*DHI*) and Direct Normal Irradiance (*DNI*) multiplied with the cosine of Sun’s zenith distance.

$$GHI = DHI + DNI \cos(z_{\odot}) \quad (1.1)$$

The focus of this paper is the second term on the right hand side of the equation (1.1). In [Ćatović & Žaknić-Ćatović 2024] this term was referred to as Direct-Beam Solar Irradiance (*DBSI*), while in [Zhang et al 2024], the same quantity was referred to as Direct Horizontal Irradiance or (*DirHI*). Further in this paper the *DBSI* term is used, noting that it is the same quantity as *DirHI*.

$$DBSI = DirHI = DNI \cos(z_{\odot}), \quad (1.2)$$

It will be shown that *DBSI* can be expanded into infinite trigonometric series completely parameterized in terms of Earth’s orbital elements, obliquity and local geographic coordinates. The *DBSI* series do not require location-specific pre-calculated empirical coefficients, normally needed when irradiance is expanded in Fourier series (e.g. [Milankovitch 1920], [Spencer 1973]), or combination of Fourier series and Legendre polynomials (e.g. [North & Caokley 1979]). Frequencies of individual terms depend on solar hour angle and its ecliptic longitude, with the amplitudes and phases depending on Earth’s orbital elements and obliquity. The series are fast converging, easy to compute and very accurate even if truncated after first several terms while providing time-continuous, globally covered solar irradiance. Importantly, the *DBSI* approximating series can easily be directly applied to studies with time-variable orbital elements or studies of irradiance at other planets in the solar system.

Unlike empirical or location-specific models, the proposed formulation is analytic, continuous, and rapidly convergent: retaining only the first three terms yields annual accuracy within $\pm 0.4 \text{ W/m}^2$ at any latitude, while four terms reduce the error to $\pm 0.04 \text{ W/m}^2$. Because the expansion depends solely on orbital geometry and a simple atmospheric transmissivity model, it is directly applicable to other planets and to paleoclimate scenarios involving time-varying orbital parameters.

The remainder of this paper is organized as follows. Section 2 introduces the parametric trigonometric expansion of the *DBSI*. Section 3 examines the convergence and accuracy of successive truncations of this expansion, benchmarked against the *DBSI* computed from high-precision astronomical ephemerides [Folkner et.al. 2014], including its accuracy across latitudes

from the equator to the poles. Section 4 validates the complete model – including two variants of atmospheric transmittance functions $\Theta_{t,\lambda,z_\odot}$ (Appendix F) – against three independent ground-based clear-sky measurement sites. Section 5 summarizes the main findings of the paper, and provides an outline of future research.

Figure 1 summarizes, as a single schematic, the computational pipeline used throughout this paper to produce and validate every *DBSI* series discussed below: the exact-geometry (‘true’) pipeline built directly from dynamical-ephemeris and measured-TSI data, the closed-form trigonometric-series approximation derived in Section 2, the shared atmospheric transmittance model of Appendix F feeding both, and the three ground-based clear-sky sites (Section 4) used to validate the combined result.

DBSIv00 simulation pipeline

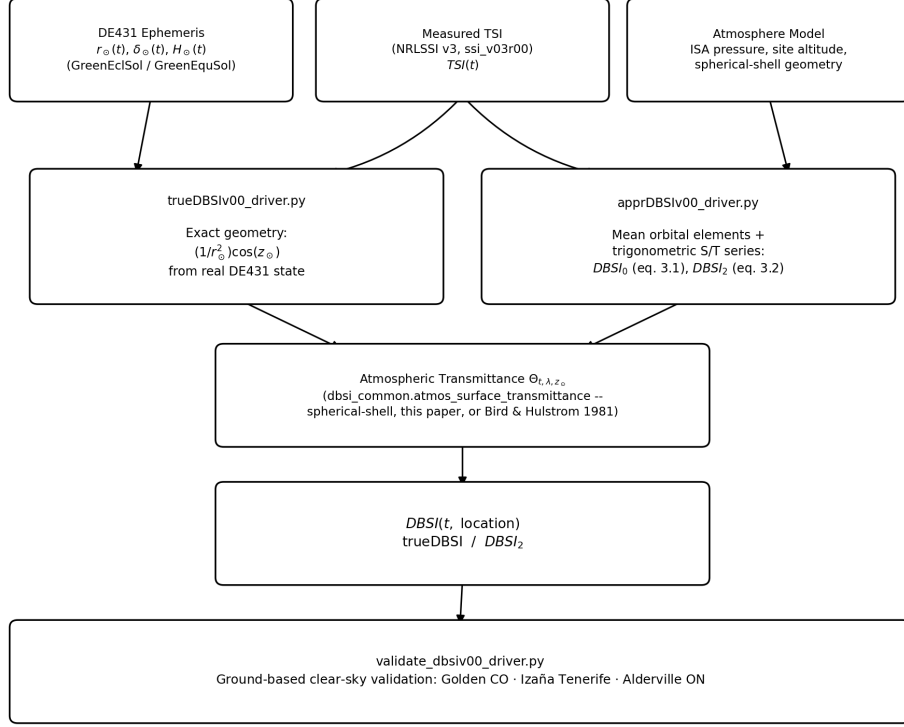


Figure 1: *Schematic of the DBSIv00 computational pipeline: independent input data sources (top), the two parallel geometry pipelines – exact (‘true’, from DE431 ephemerides) and closed-form trigonometric-series approximation (DBSI₀/DBSI₂, Section 2) – both evaluated against the same atmospheric transmittance model (Appendix F), and validated against three ground-based clear-sky sites (Section 4).*

2. Trigonometric Time Series for $DBSI$

The $DBSI_{t,\lambda}$ is the incoming solar radiation pointed directly at a specific location on Earth, after absorption and scattering out of the beam are accounted for. The $DBSI_{t,\lambda}$, as defined in (1.2), represents a particular solution of the radiation transfer equation [Chandrasekhar 1949]. In its simplest form, the solution of the radiative transfer equation can be expressed as:

$$DBSI_{t,\lambda} = DNI_{t,\lambda} \cos(z_{\odot}) = L_{t,\lambda}^{\odot} \left(\frac{R_{\odot}}{r_{\odot}} \right)^2 \Theta_{t,\lambda,z_{\odot}} \cos(z_{\odot}) \quad (2.1)$$

where $L_{t,\lambda}^{\odot}$ is the solar spectral luminosity per unit area (λ is wavelength of the outgoing radiation), $R_{\odot}$ Sun's radius, $r_{\odot}$ Sun-Earth distance, $\Theta_{t,\lambda,z_{\odot}}$ the atmospheric transmittance along the beam's path, and $z_{\odot}$ is Sun's zenith distance. The $L_{t,\lambda}^{\odot}$ accounts for solar radiation output, term $(R_{\odot}/r_{\odot})^2$ accounts for conservation of the radiant energy after it traversed the Sun-Earth distance, $\cos(z_{\odot})$ accounts for projection effect of radiation beam on the Earth's surface and finally $\Theta_{t,\lambda,z_{\odot}}$ approximates the energy losses as beam passes through the atmosphere. In this paper it is shown that the $DBSI_{t,\lambda}$ can be expressed in the the following form (for details about the derivation see Appendices):

$$DBSI_{t,\lambda} = \begin{cases} L_{t,\lambda}^{\odot} \left(\frac{R_{\odot}}{a} \right)^2 \Theta_{t,\lambda,z_{\odot}} [S_{l_{\odot}} \sin(\varphi) + T_{l_{\odot},t_{\odot}} \cos(\varphi)]; & t_{\odot} \in [t_{\odot}^{rise}, t_{\odot}^{set}] \\ 0; & t_{\odot} \in [t_{\odot}^{set}, t_{\odot}^{rise}], \end{cases} \quad (2.2)$$

Parameter a is the semi-major axis of the Earth's heliocentric orbit, φ is the geographic latitude, while $l_{\odot}$ and $t_{\odot}$ are the Sun's ecliptic longitude and local hour angle, respectively. Functions $S_{l_{\odot}}$ and $T_{l_{\odot},t_{\odot}}$ are defined as:

$$S_{l_{\odot}} = S_0 + S_1 \sin(l_{\odot} - l_1) + S_2 \sin(2l_{\odot} - \tilde{\omega}) + S_3 \sin(3l_{\odot} - 2\tilde{\omega}). \quad (2.3)$$

$$\begin{aligned} T_{l_{\odot},t_{\odot}} = & T_0 \cos(t_{\odot}) + T_1(\cos(t_{\odot} - l_{\odot} + \Phi_1) + \cos(t_{\odot} + l_{\odot} - \Phi_1)) + \\ & T_2(\cos(t_{\odot} - 2l_{\odot} + \Phi_2) + \cos(t_{\odot} + 2l_{\odot} - \Phi_2)) + \\ & T_3(\cos(t_{\odot} - 3l_{\odot} - \Phi_3) + \cos(t_{\odot} + 3l_{\odot} + \Phi_3)) + \\ & T_4(\cos(t_{\odot} + 4l_{\odot} - \Phi_4) + \cos(t_{\odot} - 4l_{\odot} + \Phi_4)) + \\ & \dots \end{aligned} \quad (2.4)$$

Table 1 shows values of amplitudes S_0, S_1, S_2, S_3 , periods and phases associated with each trigonometric terms in (2.3). The values of these parameters are calculated using the formulae presented in the Appendix E. The eccentricity was set to $e = 0.01671$, obliquity to $\varepsilon = 23.^{\circ}44$ and longitude of perihelion to $\tilde{\omega} = 103.^{\circ}16$, which are the annual mean values of these parameters for year 2022.(see [Folkner et.al. 2014]).

term	S amplitudes	S periods	S phases
0	-0.006478	n/a	n/a
1	0.398091	365. ^d 24	359 [°] .998
2	-0.006652	182. ^d 62	103 [°] .161
3	0.000028	121. ^d 75	206 [°] .322

Table 1: Calculated S-series parameters.

Table 2 shows values of amplitudes T_0, T_1, T_2, T_3, T_4 , periods and phases $\Phi_1, \Phi_2, \Phi_3, \Phi_4$ associated with the first four T-series terms. The amplitudes, periods and phases associated with the first four trigonometric terms in T-series (2.4) also depend on eccentricity e , longitude of perihelion $\tilde{\omega}$ and obliquity ε . Analytical expressions for the T-series parameters are presented in the Appendix E. The values obtained in Table 2 are calculated using the same Earth's orbital elements as in Table 1.

term	T amplitudes	T periods	T phases
0	0.959850	24. ^h 00	0 [°] .0
1	0.015732	23. ^h 93, 24. ^h 07	103 [°] .718
2	0.020580	23. ^h 87, 24. ^h 13	359 [°] .917
3	0.000348	23. ^h 80, 24. ^h 20	77 [°] .110
4	0.000223	23. ^h 74, 24. ^h 27	180 [°] .164
...	...	...	...

Table 2: Calculated T-series parameters.

In order to make effective use of expressions (2.2) – (2.4) solar hour angle $t_{\odot}$ and the solar ecliptic longitude $l_{\odot}$ must be defined in a common time scale.

Convenient choice for this time scale is local mean time t associated with perfectly uniform Earth's rotation, which differs from the mean Greenwich time (UTC) by geographic longitude l ($t=UTC+l$). The difference between the true solar time and the mean time, denoted by $E_{\odot}$ is the 'equation of time' which depends on Earth's orbital elements and obliquity (for derivation see Appendix C). The local solar hour angle can be defined as [Green 1985]:

$$t_{\odot} = \frac{2\pi}{P_d}t + l - E_{\odot}, \quad (2.5)$$

with P_d defining the period of Earth's rotation.

The equation of time $E_{\odot}$ itself admits a closed-form trigonometric expansion in Earth's orbital elements and obliquity, obtained as the sum of independent eccentricity and obliquity contributions (for the complete derivation see Appendices A and C):

$$E_{\odot} \approx 2e \sin \left(\frac{2\pi}{P_a}(t - t_p) \right) + \frac{5}{4}e^2 \sin \left(\frac{4\pi}{P_a}(t - t_p) \right) - \left[y \sin(2l_{\odot}) - \frac{y^2}{2} \sin(4l_{\odot}) + \frac{y^3}{3} \sin(6l_{\odot}) - \frac{y^4}{4} \sin(8l_{\odot}) \right],$$

with P_a and t_p as in equation (2.6) below, $y = \tan^2(\varepsilon/2)$, and $l_{\odot}$ given by that same equation; see equation (C.8) in Appendix C for the full derivation and its accuracy.

In the same time-scale the apparent solar longitude can be approximated from Kepler's laws (for derivation see Appendix A):

$$l_{\odot} \approx \frac{2\pi}{P_a}(t - t_p) + \tilde{\omega} + 2e \sin \left(\frac{2\pi}{P_a}(t - t_p) \right), \quad (2.6)$$

with P_a being period of the Earth's revolution around Sun, and t_p is a moment of Earth's passing through its perihelion. Equation (2.6) (for derivation see Appendix A) is an approximation with limited, but sufficient accuracy to the first degree of eccentricity (for orbital motions with small eccentricities). If higher accuracy is required it is relatively easy to include terms associated with higher powers of eccentricity into equation (2.6).

The harmonic form of the *DBSI* equations (2.2)–(2.4) is targeted for two reasons: (1) to precisely quantify the contributions of different harmonic

terms to solar irradiance, and (2) to prepare a monotonically decreasing, harmonic expansion of *DBSI* for future use in calculations of Land Surface Temperature (LST), where such terms naturally appear in the solution of the one-dimensional radiative heating of the surface [Ćatović et.al. 2022].

3. Convergence and Accuracy of Successive DBSI Approximations

The trigonometric expansion introduced in Section 2 provides a hierarchy of increasingly accurate approximations to the DBSI. In this section, the convergence properties of these approximations are examined and their accuracy is quantified relative to DBSI computed using high-precision astronomical ephemerides [Folkner et.al. 2014]. The goal is twofold:

- to demonstrate that the series converges rapidly for Earth-like orbital eccentricities, and
- to illustrate the geographical and seasonal dependence of the approximation error.

3.1. Convergence and Accuracy of DBSI Approximations Across Successive Orders

The rapid convergence of the DBSI series arises from the fact that the amplitudes in T-series are getting progressively smaller with their order (see Table 2 and equations (E.3)). The same can be concluded about the amplitudes of S-series, even though these series are truncated at order 4. Both sets of amplitudes are getting progressively smaller with the higher order because of the two facts (a) eccentricity for closed orbits is smaller than 1 ($e < 1$) and (b) sinus of obliquity is also smaller than 1 $\sin(\varepsilon) \leq 1$. This means that expansion (2.2) – (2.4) would converge fast for any eccentricity e , obliquity ε and longitude of perihelion $\tilde{\omega}$, which appears in both amplitudes and phases of S and T series (see E.2) – (E.4)). Since convergence is easily achieved, the truncated DBSI expansions are suitable for impact assessments of the time-varying orbital elements on the insolation. For example, it is known that Sun-Earth distance is affected by the lunar gravitational influence. As shown in Appendix B (see Figures 9 and 10) approximations of Sun Earth distance oscillate around the 'true Sun-Earth' distance.

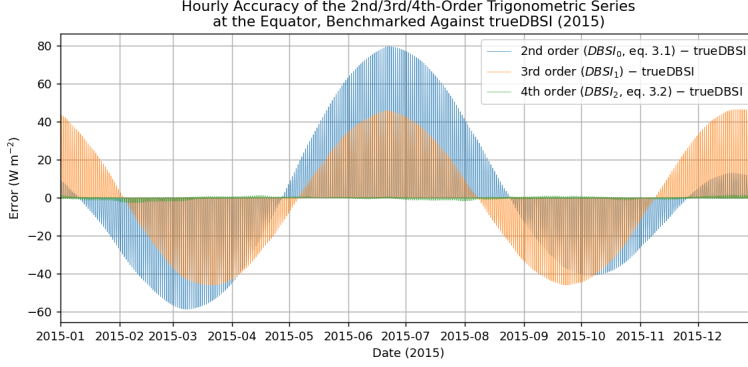


Figure 2: *Hourly accuracy of the second ($DBSI_0$, eq. 3.1), third (an intermediate truncation adding S_2 and T_1), and fourth order ($DBSI_2$, eq. 3.2) trigonometric series at the equator for year 2015, benchmarked against trueDBSI (real DE431 ephemeris and measured TSI).*

At the equator, successively adding harmonics reduces the maximum hourly error from 79.5 W/m^2 (second order, $DBSI_0$) to 46.4 W/m^2 (third order) to 2.7 W/m^2 (fourth order, $DBSI_2$) – a direct illustration of the rapid convergence discussed above. Essentially all of the residual error remaining at fourth order is attributable to the atmospheric transmittance model and the mean-vs-true orbital elements approximation (Appendix B), not to the harmonic truncation itself.

For the mean values of Earth’s orbital parameters, the DBSI series (Equation (2.2)) can be truncated after any number of terms in the $S_{l_\odot}$ and $T_{l_\odot, t_\odot}$ expansions. Two specific approximations are used to evaluate their accuracy:

- Approximation $DBSI_0$ retains the constant term S_0 and the first solar-longitude harmonic S_1 in the S-series and the leading diurnal term T_0 in the T-series. This captures the dominant annual and diurnal modulation through $\sin(l_\odot)$ and $\cos(t_\odot)$ respectively.

$$\begin{aligned}
DBSI_0 &= L_{t,\lambda}^{\odot} \left(\frac{R_{\odot}}{a} \right)^2 \times \Theta_{\lambda,z_{\odot}} \times \quad (3.1) \\
&\times \left[\mathbf{sin}(\varphi) \left[-0.006478 + 0.398091 \sin(l_{\odot} - 359.^{\circ}998) \right] + \right. \\
&\quad \left. + 0.959850 \mathbf{cos}(\varphi) \cos(t_{\odot}) \right]
\end{aligned}$$

- Approximation $DBSI_2$ extends $DBSI_0$ by adding the next two solar longitude harmonics (with amplitudes S_2 and S_3 and 6 month and 4 months periods respectively) and the next two mixed solar longitude / hour angle terms associated with amplitudes T_1 and T_2 .

$$\begin{aligned}
DBSI_2 &= DBSI_0 + L_{t,\lambda}^{\odot} \left(\frac{R_{\odot}}{a} \right)^2 \times \Theta_{\lambda,z_{\odot}} \times (3.2) \\
&\times \left[\mathbf{sin}(\varphi) \left[-0.006652 \sin(2l_{\odot} - 103.^{\circ}161) + 0.000028 \sin(3l_{\odot} - 206.^{\circ}322) \right] + \right. \\
&+ \mathbf{cos}(\varphi) \left[+0.015732 \left(\cos(t_{\odot} - l_{\odot} + 103.^{\circ}718) + \cos(t_{\odot} + l_{\odot} - 103.^{\circ}718) \right) + \right. \\
&\quad \left. \left. + 0.020580 \left(\cos(t_{\odot} - 2l_{\odot} + 359.^{\circ}918) + \cos(t_{\odot} + 2l_{\odot} - 359.^{\circ}918) \right) \right] \right]
\end{aligned}$$

The first multiplying factor in equations (3.1) – (3.2) is often referred to as a solar constant. It is not truly a constant as it varies very little over time, primarily due to changes in solar luminosity. Currently, the solar constant value is 1361 W/m^2 [Coddington et.al. 2015]. The second factor in equation (3.1) and (3.2) is the atmospheric transmissivity function $\Theta_{\lambda,z_{\odot}}$. In this work a specific form of transmissivity function (see Appendix F for more details), was used which represents a homogeneous, spherical-shell atmosphere taking into account vertically stratified density and a single, extinction cross section (no spectral phenomena were simulated for this benchmark). It is important to emphasize here that atmospheric transmissivity also depends on Earth’s orbital elements and obliquity (since Sun’s zenith distance depends on the same parameters). In the Appendix D it is shown, see formula (D.7), how Sun’s zenith distance can be expanded in series depending on its longitude and hour angle. A second, structurally richer form of $\Theta_{\lambda,z_{\odot}}$

is also used in this paper: the broadband Bird & Hulstrom (1981) model [Bird & Hulstrom 1981], which factors atmospheric attenuation into five separate physical processes (Rayleigh scattering, ozone absorption, uniformly-mixed-gas absorption, water-vapor absorption and aerosol extinction) rather than the single lumped extinction cross section above. Both forms of $\Theta_{\lambda, z_{\odot}}$ are compared directly against ground-based measurements in Section 4 and Appendix F.2.

3.2. Accuracy of DBSI Approximations at Different Latitudes

The *DBSI* approximations provide global coverage, i.e., they are valid for any latitude from the equator to the poles. Figure 3 shows calculated $DBSI_2$ (eq. 3.2) at local solar noon for latitudes of 0° , $\pm 15^\circ$, $\pm 45^\circ$, $\pm 75^\circ$, and $\pm 90^\circ$ (top row), together with the corresponding error against trueDBSI – the same real-ephemeris, measured-TSI reference used throughout this section – at each latitude (bottom row). Errors do not show a strong latitude dependence: maximum error ranges from about 2.0 W/m^2 near the poles to about 3.4 W/m^2 near mid-latitudes ($\pm 45^\circ$), with the equator falling in between (2.6 W/m^2). This total error combines the harmonic-truncation error quantified in Figure 2 above (sub- 3 W/m^2 for $DBSI_2$ alone) with the smaller mean-vs-true orbital elements error already discussed in Appendix B (of order $1\text{--}2 \text{ W/m}^2$). The approximation therefore provides consistently sub- 4 W/m^2 accuracy across the full range of latitudes from equator to pole.

Simplicity, global coverage and arbitrary accuracy make the approximation (2) suitable for various solar applications, calculations of surface temperatures (see [Ćatović & Žaknić-Ćatović 2024]) and parametric climate sensitivity assessments.

Time-continuity feature of DBSI approximations is demonstrated in Figure 4, where annual DBSI approximation (3.2) at latitude $\varphi = 43^\circ.65N$ - Toronto, Canada) is shown.

4. Validation Against Ground-Based Clear-Sky Measurements

The comparisons in Section 3 benchmark the DBSI trigonometric series against itself, computed from dynamical ephemerides – they isolate the accuracy of the geometric $(1/r_{\odot})^2 \cos(z_{\odot})$ expansion from the atmospheric trans-

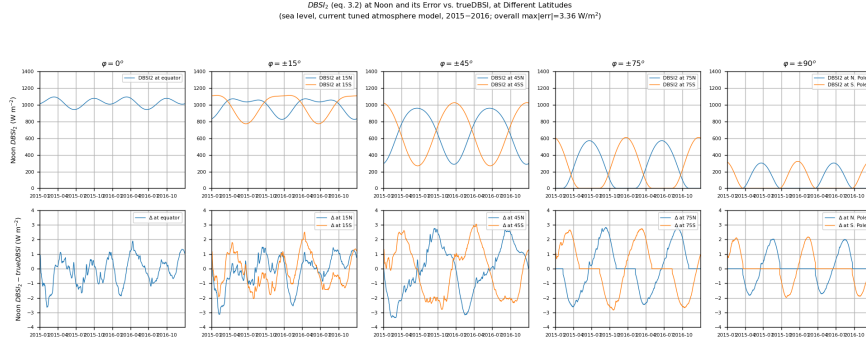


Figure 3: Noon $DBSI_2$ (eq. 3.2, top row) and its error against trueDBSI (bottom row) at nine latitudes from equator to pole, 2015–2016, current tuned atmosphere model (Appendix F.1).

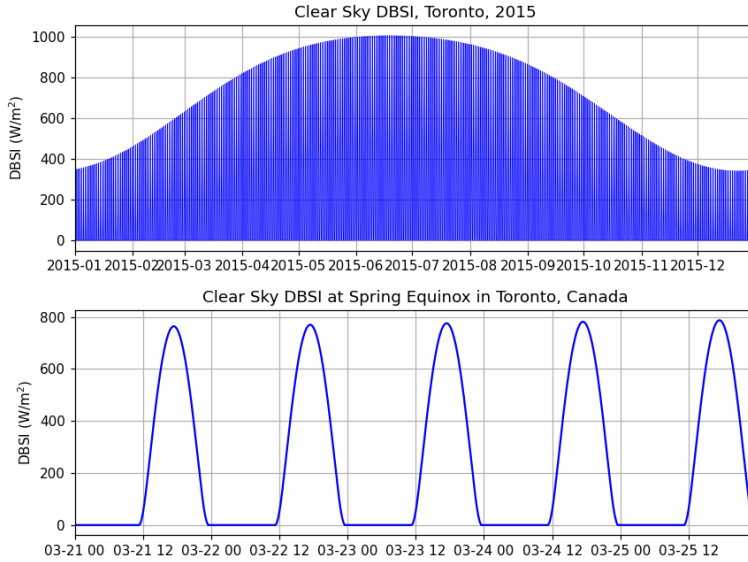


Figure 4: Annual and daily variability of DBSI, under clear-sky conditions, in Toronto.

mittance factor $\Theta_{t,\lambda,z_\odot}$. To assess the complete model, including $\Theta_{t,\lambda,z_\odot}$, against real measurements, three independent clear-sky ground stations were used: Golden, Colorado (NREL, 2024-03-09, a Global–Diffuse target that isolates direct-beam irradiance, matching what DBSI is defined to represent), Alderville, Ontario (2015-03-24, a horizontal pyranometer network [Gagné et al. 2016] whose reading includes diffuse irradiance the model does not produce), and Izaña, Tenerife (BSRN station, 2010-05-16, independently confirmed clear-sky by [García et al. 2013] – its pyrheliometer records true direct-normal irradiance, projected here onto the horizontal via this pipeline’s own modeled zenith distance, since the BSRN archive does not carry an independently measured zenith angle for this station). Alderville’s residuals therefore combine a genuine transmittance mismatch with an unavoidable, roughly constant diffuse offset, and are reported for completeness rather than as a primary accuracy benchmark; Golden and Izaña, both direct-beam-only targets, are the cleaner comparisons throughout this section. Izaña’s elevation (2,373 m, versus Golden’s $\sim 1,730$ m and Alderville’s 259 m) also exercises the altitude-aware spherical-shell transmittance model (Appendix F) at a substantially higher point than either of the other two sites.

The atmospheric-transmittance step uses a single, common parameter set at every site: atmosphere thickness $H_a = 100$ km, surface density $\varrho_0 = 1.22$ kg/m³, and extinction cross section $\sigma_{\text{ext}} = 0.00250$ m²/kg (tuned against a pooled Golden+Izaña objective, Appendix F.1) – identical for Golden, Alderville and Izaña, and used to produce Figures 5–7 and Table 3 below. Only each site’s own elevation enters differently, through the altitude-aware evaluation of the spherical-shell model (Appendix F) at that site’s actual altitude – Golden ($\sim 1,730$ m), Alderville (259 m), Izaña (2,373 m) – rather than through any per-site re-tuning of the atmosphere’s extinction, thickness or density.

The purely geometric part of the model was validated independently of $\Theta_{t,\lambda,z_\odot}$ by comparing the modeled solar zenith distance at Golden against the station’s own measured value: RMSE = 0.09° over the full clear-sky day, confirming the longitude-aware hour angle, time-zone conversion, and declination chain (Sections 2 and 4, Appendices A–D) are correct well beyond the resolution DBSI needs. Every residual reported below is therefore attributable to

$\Theta_{t,\lambda,z_\odot}$, not to the orbital/geometric expansion that is this paper’s primary subject.

Table 3 compares two forms of $\Theta_{t,\lambda,z_\odot}$ against all three sites, for the full (‘true’) *DBSI* and the four-term truncated series *DBSI*₂ of equation (3.2): the single-extinction-cross-section spherical-shell model of Appendix F, with its one free parameter tuned as described in Appendix F.1 against a pooled Golden+Izaña objective, and the broadband Bird & Hulstrom (1981) model [Bird & Hulstrom 1981], with its one free parameter tuned as described in Appendix F.2 against Golden only – each applied unmodified (Alderville included) at whichever site or sites it was not tuned on.

site	model	RMSE (W/m ²)	bias (W/m ²)	max err (W/m ²)
Golden	Θ (this paper), <i>DBSI</i>	36.0	+22.0	66.8
Golden	Θ (this paper), <i>DBSI</i> ₂	33.9	+19.4	63.4
Golden	Bird & Hulstrom, <i>DBSI</i>	7.1	+2.8	15.8
Alderville	Θ (this paper), <i>DBSI</i>	30.2	−24.9	60.0
Alderville	Θ (this paper), <i>DBSI</i> ₂	32.0	−26.9	62.2
Alderville	Bird & Hulstrom, <i>DBSI</i>	54.8	−49.2	76.4
Izaña	Θ (this paper), <i>DBSI</i>	21.3	−14.9	42.6
Izaña	Θ (this paper), <i>DBSI</i> ₂	21.7	−15.2	43.4
Izaña	Bird & Hulstrom, <i>DBSI</i>	57.6	−38.2	116.2

Table 3: *DBSI* validation against ground-based clear-sky measurements, after independently tuning each transmittance model’s own free parameter – this paper’s Θ against a pooled Golden+Izaña objective, Bird & Hulstrom against Golden only (Appendix F.1–F.2). Golden’s and Izaña’s targets isolate direct-beam irradiance; Alderville’s includes diffuse (see text).

Figures 5–7 show all three comparisons in full: measured irradiance against both *DBSI* forms and the Bird & Hulstrom (1981) alternative (top panels), and the corresponding model-minus-measured residual through the day (bottom panels).

Three observations follow. First, at both direct-beam-only sites the truncated *DBSI*₂ reproduces the full *DBSI* to within 0.4–2.1 W/m² RMSE, confirming that the harmonic-truncation error quantified in Section 3 (sub-W/m²) remains small next to the residual attributable to atmospheric transmittance (7–58 W/m², even after tuning). This is a practical demonstration of the point made in Section 1: a well-converged geometric expansion is, in a real clear-sky comparison, ultimately limited by the transmittance model, not

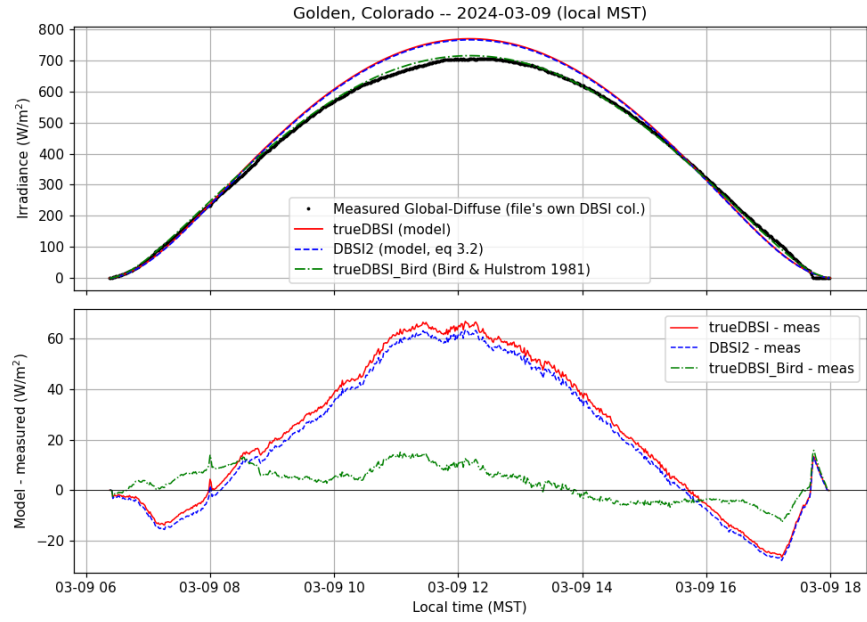


Figure 5: *DBSI validated against ground-based clear-sky measurements at Golden, Colorado. Top: measured irradiance vs. this paper's transmittance model (DBSI, DBSI₂, atmosphere parameters per Appendix F.1) and the Bird & Hulstrom (1981) alternative. Bottom: model – measured residual through the day.*

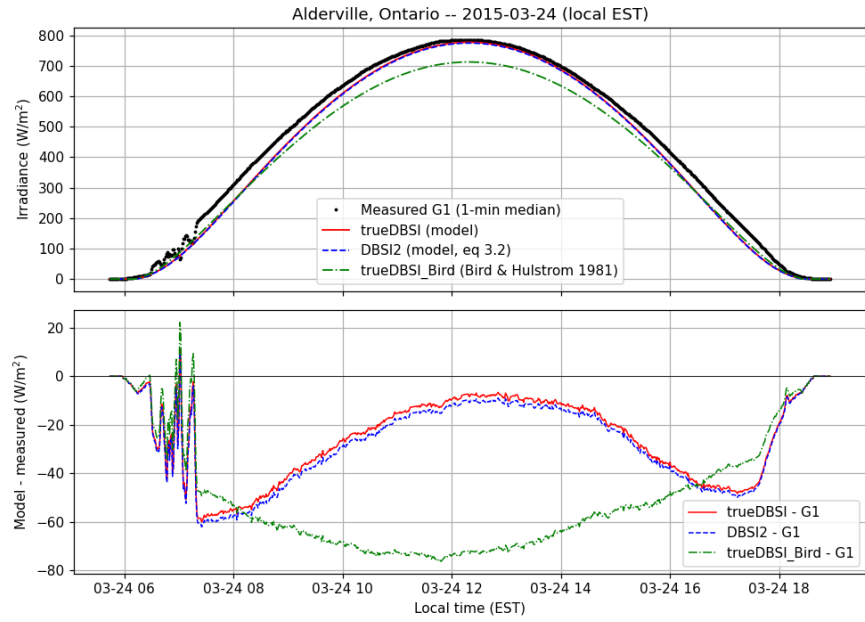


Figure 6: *DBSI validated against ground-based clear-sky measurements at Alderville, Ontario (diffuse-inclusive target – see text). Top: measured irradiance vs. this paper’s transmittance model (DBSI, $DBSI_2$, same atmosphere parameters as Golden and Izaña – Appendix F.1) and the Bird & Hulstrom (1981) alternative. Bottom: model – measured residual through the day.*

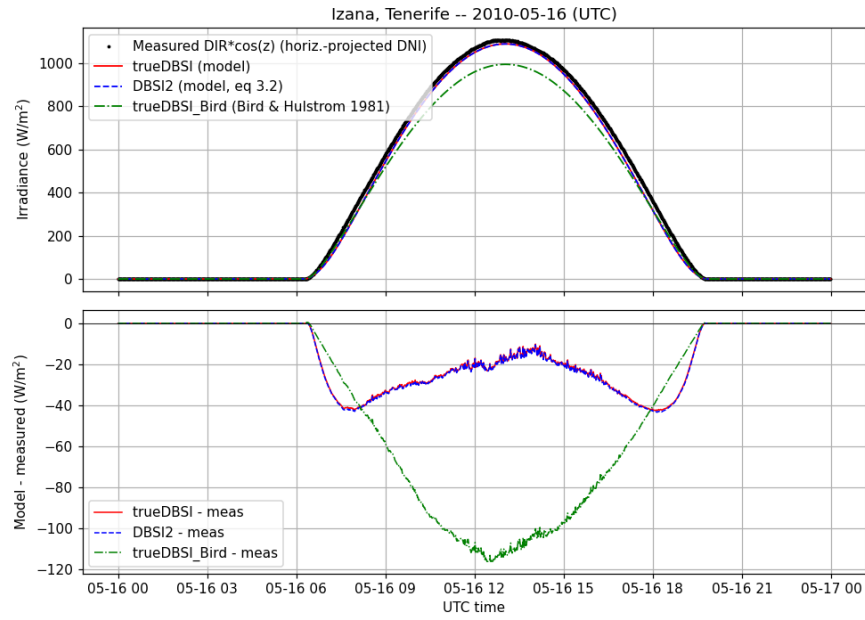


Figure 7: *DBSI validated against ground-based clear-sky measurements at Izaña, Tenerife. Top: measured irradiance vs. this paper’s transmittance model (DBSI, DBSI₂, same atmosphere parameters as Golden and Alderville – Appendix F.1, evaluated at Izaña’s own altitude) and the Bird & Hulstrom (1981) alternative. Bottom: model – measured residual through the day.*

by the harmonic truncation. Second, because σ_{ext} is now tuned against a *pooled* Golden+Izaña objective rather than Golden alone (Appendix F.1), this paper’s Θ no longer wins at every site: at Golden specifically, the structurally richer, five-process Bird & Hulstrom (1981) model – still tuned against Golden only – now outperforms this paper’s Θ by a factor of ~ 5 , and the residual’s shape changes qualitatively: Figure 5 shows this paper’s Θ residual is now a broad, mostly *positive* plateau through the day (roughly $+55$ to $+67 \text{ W/m}^2$ near midday, falling to small negative values near sunrise), reflecting that $\sigma_{\text{ext}} = 0.00250 \text{ m}^2/\text{kg}$ under-attenuates relative to Golden’s own single-site optimum ($0.00295 \text{ m}^2/\text{kg}$), while the Bird & Hulstrom residual, still fit to Golden alone, remains flat and small. Third, that trade pays off at the other two sites: this paper’s Θ now clearly outperforms Bird & Hulstrom’s Golden-tuned turbidity at both Alderville (RMSE 30.2 vs. 54.8 W/m^2) and, most notably, Izaña (RMSE 21.3 vs. 57.6 W/m^2) – consistent with Bird & Hulstrom’s one free parameter having been tuned against Golden’s more turbid, lower-altitude atmosphere and not transferring to either site’s cleaner air, whereas σ_{ext} ’s pooled Golden+Izaña tuning explicitly trades away some of Golden’s individual accuracy for a value that generalizes better across sites. Either way, a single site- or pair-tuned parameter is not a substitute for the spectrally- and altitude-resolved treatment left to the companion paper [Ćatović & Žaknić-Ćatović 2026]. Appendix F.1–F.2 discusses the tuning methodology behind Table 3 and why the residual’s shape, not just the parameter count, is the more informative result.

5. Summary and Conclusion

The following important observations can be made about the DBSI model (2.2) – (2.6):

- The series converge fast, especially for small eccentricities. If truncated after the first term T_1 approximated irradiance has accuracy (relative to *DBSI* calculated with ‘exact’ Sun-Earth distance factor and solar declination) of $\pm 44 \text{ W/m}^2$ after second term T_2 approximated irradiance have accuracy of $\pm 1.2 \text{ W/m}^2$, while with the T_3 term included accuracy is $\pm 0.4 \text{ W/m}^2$. Finally with the fourth term T_4 included accuracy

is $\pm 0.04 \text{ W/m}^2$. (These figures isolate pure harmonic-truncation error against a mean-element 'exact' reference; see Section 3 for the corresponding benchmark against real DE431 ephemerides and measured TSI, which additionally includes the mean-vs-true orbital-element error -2.7 W/m^2 at the equator, up to 3.4 W/m^2 across latitudes, for the same fourth-order truncation.)

- Precision of the series depends on geographic latitude and time. At poles *DBSI* approximation depend only on solar longitude and is most precise, while at equator it depends only on solar hour angle, and is least precise.
- The series are time-continuous and have global coverage, i.e. the same expression can be used for any point on the Earth's surface.

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Code and Data Availability

A companion Python implementation of the closed-form $DBSI_0$ and $DBSI_2$ approximations derived in this paper, together with a small graphical interface for exploring them at any location, date/time range, and set of orbital elements, is publicly available at <https://github.com/catovicone-gif/DBSI-Explorer> under the MIT license. This repository is a focused, coarse demonstration of

the approximations presented here; it does not include the larger validation and sensitivity-analysis codebase used to produce this paper’s figures and tables.

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Appendices

Process followed to construct the functions $S_{l_{\odot}}$ and $T_{l_{\odot}, t_{\odot}}$, in terms of Earth's orbital elements and obliquity is laid out in the appendices. In Appendix A approximations of solar longitude, in terms of local time, is discussed. Appendix B derives approximations for the Sun-Earth distance factor and provides comparisons with the currently popular approximations. Appendix C derives a closed-form, fast-converging trigonometric expansion for the solar hour angle and the equation of time $E_{\odot}$ used in the definition of the local solar hour angle $t_{\odot}$, following [Lahaye 2012, Meeus 1998]. In Appendix D approximation of the solar declination as a function of the longitude, is discussed. Appendix E provides derivation of function $S_{l_{\odot}}$ and first four terms of the function $T_{l_{\odot}, t_{\odot}}$. Finally Appendix F, provides discussion of the atmospheric transmittance function used in this paper. Since the transmittance itself is the most important component of and model approximating the irradiance on Earth's surface, a more detailed discussion of this function will be provided in a separate paper [Ćatović & Žaknić-Ćatović 2026].

A. The Time Approximation of the Sun's Longitude $l_{\odot}$ Using Kepler laws

Assuming elliptical orbit of Earth around the Sun (the first Kepler's law), the heliocentric Sun-Earth distance ($r_{\oplus}$) or the geocentric Sun-Earth distance denoted by $r_{\odot} = r_{\oplus}$, can be expressed as:

$$r_{\oplus} = \frac{a(1 - e^2)}{1 + e \cos(l_{\oplus} - \tilde{\omega})} = r_{\odot} = \frac{a(1 - e^2)}{1 - e \cos(l_{\odot} - \tilde{\omega})}. \quad (\text{A.1})$$

The angle appearing in cosine functions of (A.1) is known as true anomaly ϑ , [Green 1985], and can be expressed in terms of heliocentric Earth's longitude $l_{\oplus}$ (angle between Earth's position and Spring equinox), or apparent Sun's geocentric longitude $l_{\odot}$. Angle $\tilde{\omega}$ is the longitude of perihelion, it is one of the six Earth's orbital elements.

$$\vartheta = l_{\oplus} - \tilde{\omega} = l_{\odot} - \tilde{\omega} - 180^{\circ}. \quad (\text{A.2})$$

The second Kepler's law, provides relationship between the solar longitude and time:

$$\frac{1}{2}r_{\oplus}^2 \frac{dl_{\oplus}}{dt} = \frac{\pi a^2 \sqrt{(1-e^2)}}{P_a}. \quad (\text{A.3})$$

Inserting $r_{\oplus}$ from (A.1) into equation (A.3), and expanding the left hand side into power series (in powers of eccentricity) yields:

$$(1 - 2e \cos(l_{\oplus} - \tilde{\omega}) + 3e^2 \cos^2(l_{\oplus} - \tilde{\omega}) - 4e^3 \cos^3(l_{\oplus} - \tilde{\omega}) + \dots) dl_{\oplus} = \frac{2\pi}{(1-e^2)^{3/2} P_a} dt. \quad (\text{A.4})$$

For orbits with small eccentricities, as is the case with Earth, it is sufficient to retain only the first power of eccentricity. After integrating:

$$l_{\oplus} = \frac{2\pi}{(1-e^2)^{3/2} P_a} (t - t_P) + \tilde{\omega} + 2e \sin(l_{\oplus} - \tilde{\omega}), \quad (\text{A.5})$$

where t_P is the time of passing through perihelion (see figure below). Equation (A.5) is transcendental and can't be used to find the longitude directly. However, an approximation in which higher eccentricity power e^2 is neglected and $l_{\oplus} - \tilde{\omega}$ appearing in sin function is replaced by $(t - t_P)$ provides sufficiently accurate estimate $l_{\oplus}$ (or $l_{\odot}$ considering (A.2)). This approximation of Earth's or Sun's longitude was used by Milankovitch in [Milankovitch 1920] and was found sufficiently accurate for calculating longitude over very long time periods.

$$l_{\oplus} = l_{\odot} - 180^\circ = \frac{2\pi}{P_a} (t - t_P) + \tilde{\omega} + 2e \sin(t - t_P), \quad (\text{A.6})$$

For higher eccentricities, the same approach can be easily modified to include higher eccentricity powers of equation (A.4).

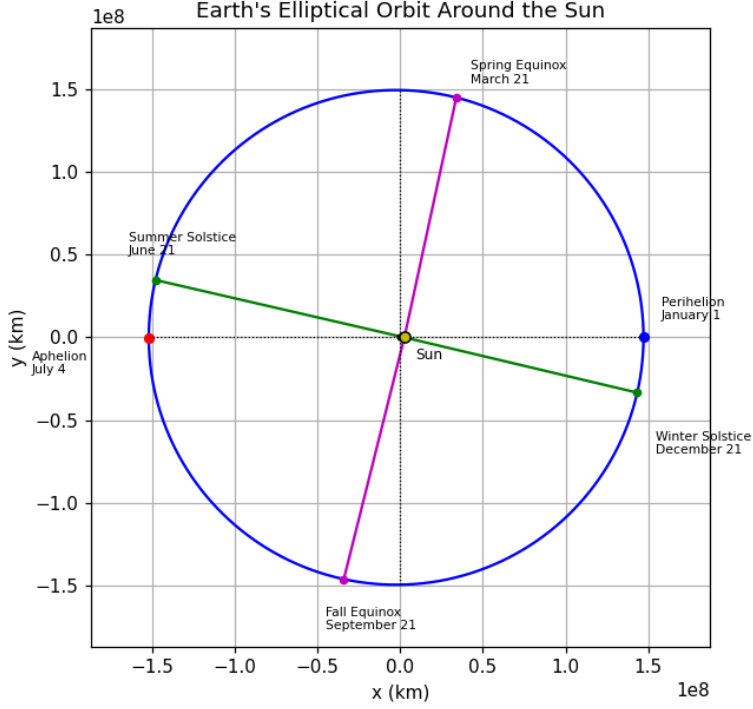


Figure 8: *Earth's Elliptical Orbit Around the Sun.*

B. Parametrization of Sun-Earth Distance Factor

In this section the Sun-Earth distance factor from (2.1), is expressed in a form which will be proven useful for the task of expanding the DBSI into infinite series. Considering the (A.1), the Sun-Earth distance factor, from equation (2.1), can be expressed as:

$$\left(\frac{1}{r_{\odot}}\right)^2 = \left(\frac{1}{a}\right)^2 [E_0 - E_1 \cos(l_{\odot} - \tilde{\omega}) + E_2 \cos(2(l_{\odot} - \tilde{\omega}))], \quad (\text{B.1})$$

with parameters E_0, E_1, E_2 dependent on orbital eccentricity:

$$E_0 = \frac{1 + \frac{e^2}{2}}{(1 - e^2)^2}, E_1 = \frac{2e}{(1 - e^2)^2}, E_2 = \frac{e^2}{2(1 - e^2)^2}. \quad (\text{B.2})$$

Equation (B.1) shows that the Sun-Earth distance factor, even in the case of unperturbed elliptical orbit (or instant - osculatory orbit), is a periodic function in solar longitude with two terms having frequencies corresponding to one year and half a year periods, while amplitudes are proportional to e and e^2 respectively. Combined with the approximation for Sun's longitude (A.6) the equation (B.1) provides a simple procedure to calculate the Sun-Earth distance factor in a local time scale. Note that the Sun-Earth distance factor depends directly on three orbital elements $a, e, \tilde{\omega}$, while it implicitly depends on the specific time-scale via the time of passing through perihelion t_P , and revolution period P_a , which appear in the solar longitude approximation (A.6). This approximation is sufficiently accurate for small eccentricities (as is the case for Earth's orbit), and can easily be expanded to higher degrees in eccentricity if needed.

The Sun-Earth distance factor expanded into Fourier series in solar longitude [Spencer 1973] (see equation (B.3)) is frequently used by the solar radiation modelers.

$$\begin{aligned} \left(\frac{1}{r_{\odot}}\right)^2 &= 1.000110 + 0.034221 \cos(\Gamma) + 0.001280 \sin(\Gamma) + \\ &\quad + 0.000719 \cos(2\Gamma) + 0.000077 \sin(2\Gamma), \\ \Gamma &= \frac{2\pi(d-1)}{365}, \quad d = 1 \dots 365 \end{aligned} \quad (B.3)$$

Another popular formula used by the modelers is an expression presented in [Duffie & Beckman 1980] or [Iqbal 1983] (see equation (B.4)).

$$\left(\frac{1}{r_{\odot}}\right)^2 = 1 + 0.033 \cos\left(\frac{2\pi d}{365}\right), \quad d = 1 \dots 365 \quad (B.4)$$

In expressions (B.3) and (B.4), Sun-Earth distance is measured in Astronomical Units.

In the remainder of this section the approximations (B.1), (B.3) and (B.4) are compared with the accurate $1/r_{\odot}^2$ obtained from dynamical ephemerides DE431 [Folkner et.al. 2014]. This DE431 reference distance is the Sun's position relative to Earth's center, taken directly from the full n-body ephemeris

integration — i.e. a heliocentric-equivalent quantity, not one referenced to the solar-system barycenter. A barycentric reference frame is neither necessary nor appropriate for this purpose: a two-body separation vector such as $r_{\odot}$ is independent of the coordinate origin used by the underlying ephemeris, and reconstructing it as the difference of two separately-referenced barycentric position vectors would risk introducing spurious error, from the non-collinearity of the Sun’s and Earth’s own barycentric position vectors, without any accuracy benefit. Hourly Sun-Earth distances $r_{\odot}$ were downloaded in the time domain from January 01, 2024 to January 01, 2025. Earth’s orbital elements needed for the (B.1)/(A.6), are mean values of orbital elements in year 2022 - also from DE431, $a = 149600438.4714$ km, $e = 0.0167141$, $\tilde{\omega} = 103.^{\circ}1609$, while perihelion passing time is set, somewhat arbitrarily, to 9:00am, January 03, 2024 – this choice of reference epoch has no bearing on the model’s accuracy, since e and $\tilde{\omega}$ change negligibly between 2022 and 2024 and a mean-element model’s error depends only on elapsed time since perihelion, not on which calendar year’s mean values are nominally used. Figure 9 (top) shows the the Sun-Earth distance, as calculated by different approaches discussed above while the same figure (bottom) shows the errors of the three methods relative to the heliocentric DE431 Sun-Earth distance. It is noticeable that the error associated with the approximation (B.1) stays within $[-41,000\text{km}, 45,000\text{km}]$ range, while error associated with the [Spencer 1973] and [Iqbal 1983] are contained within $[-64,000\text{km}, 79,000\text{km}]$ and $[-206,000\text{km}, 147,000\text{km}]$ respectively. ‘Wavy’ structure of the errors indicates the effects of Earth’s center of gravity moving around the Earth-Moon barycenter, which follows the lunar month; a Fourier analysis of the residual confirms its dominant structure sits in the 11–17 cycles/year band (the ~ 13 -cycle/year, 27–30 day lunar period), with amplitude up to 5,000km. Finally, Figure 10, shows the Sun-Earth distance factors calculated by the four methods (top graph) and their errors relative to the heliocentric DE431 distance. The error of approximation (B.1) oscillates in $[-0.06\%, 0.06\%]$ range, while [Spencer 1973] and [Iqbal 1983] relative errors oscillate in $[-0.10\%, 0.09\%]$ and $[-0.20\%, 0.28\%]$ respectively.

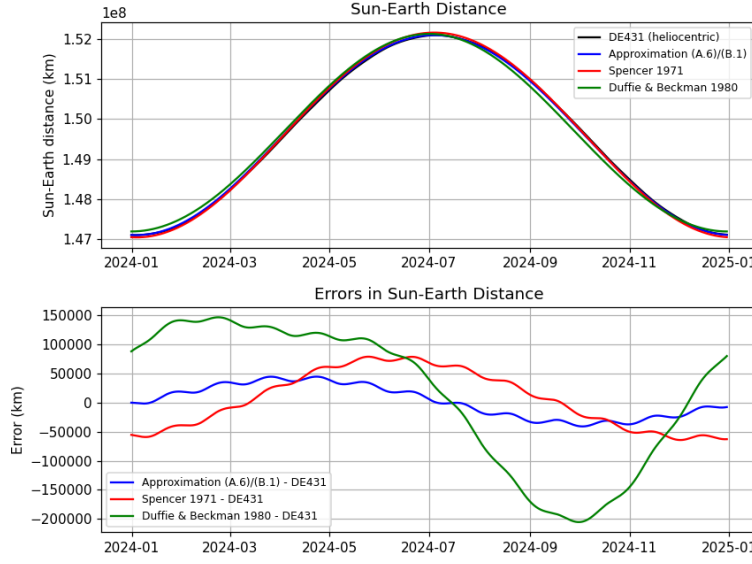


Figure 9: *Accuracy Test of Different Approximations of Sun-Earth Distance*

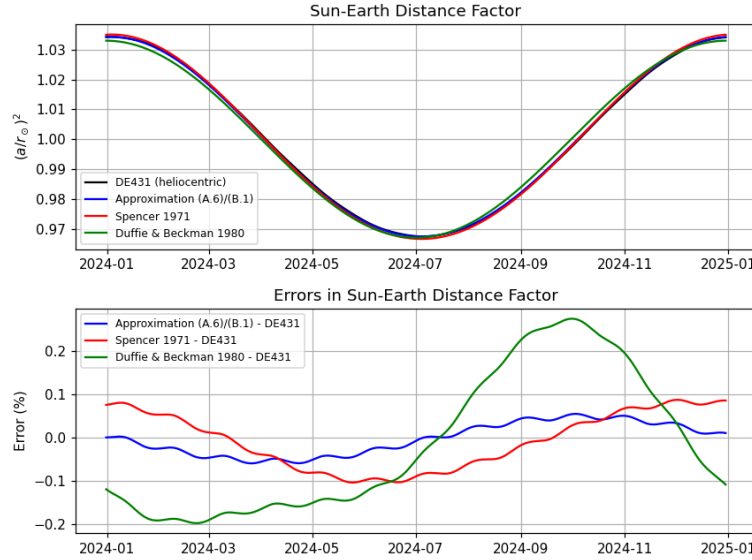


Figure 10: *Benchmarking of Different Approximations of of Sun-Earth Factor*

C. Solar Hour Angle and the Equation of Time Parametrization

In equation (2.5) the local solar hour angle $t_{\odot}$ is related to local mean time t through the equation of time $E_{\odot}$, which in this paper has so far been sourced

from dynamical ephemerides (DE431). Since $E_{\odot}$ depends, like every other quantity in Sections 2–4, only on the Earth’s orbital elements and obliquity, it admits a closed-form, fast-converging trigonometric expansion of the same character as the $S_{l_{\odot}}$ and $T_{l_{\odot}, t_{\odot}}$ series. This appendix derives that expansion, so that the DBSI model can be evaluated for any epoch or any planetary body without recourse to pre-tabulated ephemerides.

The decomposition strategy used below — splitting $E_{\odot}$ into independent eccentricity and obliquity contributions and combining them additively — follows the approach of [Lahaye 2012], extended here (i) to second order in eccentricity, using the standard equation-of-center series (see e.g. [Meeus 1998]), and (ii) to an arbitrary number of obliquity terms via a closed-form generating-function derivation (Section C.3 below), rather than the first-order truncation given in [Lahaye 2012].

C.1 Definition and decomposition

The equation of time is the difference, expressed as an hour angle, between the local mean solar time and the true (apparent) solar time. Consistent with the convention adopted in equation (2.5), $E_{\odot}$ is defined such that it is subtracted from the mean hour angle to yield the true hour angle $t_{\odot}$ used throughout this paper.

Physically, $E_{\odot}$ has two independent origins:

- (i) the *eccentricity* of Earth’s orbit, which causes the true Sun to move at a non-uniform rate along the ecliptic while the fictitious *mean Sun* advances uniformly (Kepler’s second law);
- (ii) the *obliquity* of the Earth’s rotation axis, which causes the true Sun’s right ascension to advance non-uniformly along the celestial equator even if its motion along the ecliptic were uniform, since the ecliptic and the equator are inclined by ε .

Because both effects vanish independently when $e = 0$ and $\varepsilon = 0$ respectively, $E_{\odot}$ can be obtained, to excellent accuracy, as the sum of the two contributions computed separately:

$$E_{\odot} = E_{\text{ecc}} - E_{\text{obl}}, \quad (\text{C.1})$$

with E_{ecc} and E_{obl} derived below in terms of the mean anomaly M and the solar longitude $l_{\odot}$ already defined in equations (2.6) and (A.6).

C.2 Eccentricity contribution

The eccentricity contribution is precisely the deviation of the true solar longitude from the mean longitude, i.e. the *equation of center*, already partially derived in Appendix A. Continuing the expansion of equation (A.4) one order further in e (a standard result of the series inversion of Kepler's equation, see e.g. [Meeus 1998]) gives:

$$\nu - M = 2e \sin(M) + \frac{5}{4}e^2 \sin(2M) + \mathcal{O}(e^3), \quad (\text{C.2})$$

where $M = \frac{2\pi}{P_a}(t - t_P)$ is the mean anomaly and ν the true anomaly, so that $l_{\odot} - (M + \tilde{\omega}) = \nu - M$. This is simply equation (A.6) carried to second order in e ; the first-order term $2e \sin(M)$ used in equation (A.6) is recovered by truncating (C.2) at its leading term. The eccentricity contribution to the equation of time is then

$$E_{\text{ecc}} = 2e \sin(M) + \frac{5}{4}e^2 \sin(2M). \quad (\text{C.3})$$

C.3 Obliquity contribution

The obliquity contribution is the difference between the solar longitude $l_{\odot}$ and the corresponding right ascension α , i.e. the *reduction to the equator* (cf. the analogous first-order treatment in [Lahaye 2012] and the spherical-trigonometry relations underlying equation (D.1) of this paper, following e.g. [Green 1985]). From spherical trigonometry, the right ascension of a point on the ecliptic with longitude $l_{\odot}$ satisfies

$$\tan \alpha = \cos \varepsilon \tan l_{\odot}. \quad (\text{C.4})$$

Writing $k = \cos \varepsilon$ and $u \equiv l_{\odot} - \alpha$, equation (C.4) can be solved for $\tan u$ exactly:

$$\tan u = \frac{(1 - k) \tan l_{\odot}}{1 + k \tan^2 l_{\odot}} = \frac{y \sin(2l_{\odot})}{1 + y \cos(2l_{\odot})}, \quad y \equiv \tan^2\left(\frac{\varepsilon}{2}\right) = \frac{1 - k}{1 + k}. \quad (\text{C.5})$$

Equation (C.5) is of the form $\tan u = y \sin \theta / (1 + y \cos \theta)$ with $\theta = 2l_\odot$, which is recognized as the imaginary part of $\ln(1 + ye^{i\theta})$. Expanding the logarithm for $|y| < 1$,

$$\ln(1 + ye^{i\theta}) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} y^n e^{in\theta}, \quad (\text{C.6})$$

and taking the imaginary part gives $u = \arg(1 + ye^{i\theta})$ directly as a convergent trigonometric series (no branch ambiguity, since $y < 1$ for any $\varepsilon < 90^\circ$):

$$E_{\text{obl}} \equiv l_\odot - \alpha = y \sin(2l_\odot) - \frac{y^2}{2} \sin(4l_\odot) + \frac{y^3}{3} \sin(6l_\odot) - \frac{y^4}{4} \sin(8l_\odot) + \dots \quad (\text{C.7})$$

Since $y = \tan^2(\varepsilon/2) \approx 0.0430$ for $\varepsilon = 23^\circ.44$, successive terms shrink by a factor of order $y \approx 1/23$, so the series converges very rapidly — a direct obliquity analogue of the eccentricity-power convergence already exploited for the S - and T -series in Appendix E.

C.4 Combined expansion

Collecting equations (C.1), (C.3) and (C.7), and using $l_\odot$ from equation (2.6), the equation of time takes the fully parametrized form

$$E_\odot \approx 2e \sin(M) + \frac{5}{4}e^2 \sin(2M) - \left[y \sin(2l_\odot) - \frac{y^2}{2} \sin(4l_\odot) + \frac{y^3}{3} \sin(6l_\odot) - \frac{y^4}{4} \sin(8l_\odot) \right] \quad (\text{C.8})$$

with $M = \frac{2\pi}{P_a}(t - t_P)$, $y = \tan^2(\varepsilon/2)$, and $l_\odot$ given by equation (2.6). No parameters beyond e , ε , $\tilde{\omega}$ and t_P — all already required elsewhere in this paper — are needed; equation (C.8) can therefore replace the DE431 lookup used for $E_\odot$ in equation (2.5) with an explicit analytic expression valid for any epoch and, subject to substituting the appropriate orbital elements, any planetary body. Table C.1 lists the numerical values of the coefficients in (C.8) for the same mean orbital elements used throughout this paper ($e = 0.0167141$, $\varepsilon = 23^\circ.44$, $\tilde{\omega} = 103^\circ.1609$).

C.5 Accuracy

Equation (C.8) was benchmarked against $E_\odot$ computed from an exact two-body solution of Kepler's equation (eccentric anomaly solved to machine precision, true longitude and right ascension computed exactly from e , ε , $\tilde{\omega}$

term	type	amplitude	argument
e	eccentricity	$2e = 0.033428$	M
e^2	eccentricity	$\frac{5}{4}e^2 = 0.000350$	$2M$
y	obliquity	$y = 0.043037$	$2l_\odot$
y^2	obliquity	$y^2/2 = 0.000926$	$4l_\odot$
y^3	obliquity	$y^3/3 = 0.0000266$	$6l_\odot$
y^4	obliquity	$y^4/4 = 8.6 \times 10^{-7}$	$8l_\odot$

Table 4: Amplitudes of the $E_\odot$ series, equation (C.8), for the mean 2022 orbital elements used elsewhere in this paper.

with no series truncation), over one full anomalistic year. Table C.2 shows the maximum absolute error as successive terms of (C.8) are retained.

truncation	max. error
1st-order e , 1 obliquity term	17.8 s
1st-order e , 2 obliquity terms	5.1 s
2nd-order e , 2 obliquity terms	0.44 s
2nd-order e , 3 obliquity terms	0.09 s
2nd-order e , 4 obliquity terms	0.08 s

Table 5: Maximum annual error of $E_\odot$ from equation (C.8), relative to an exact two-body Kepler solution, as a function of truncation order. Beyond the 2nd-order eccentricity / 3-term obliquity combination, the residual error is dominated by the $\mathcal{O}(e^3)$ term omitted from (C.2), not by the obliquity series.

Retaining the full expression (C.8) (2nd-order eccentricity, four obliquity terms) gives sub-second accuracy over an entire year — comparable to, or better than, the precision meaningful for the $\pm 0.4 \text{ W/m}^2$ -level DBSI truncation accuracy discussed in Section 3, since a 0.1–1 s error in $E_\odot$ corresponds to a hour-angle error of order 10^{-3} to 10^{-2} degrees, well below the resolution of the DBSI series truncated at the T_2 – T_3 level.

[Placeholder: insert a figure here, analogous to Figures 5–7, comparing $E_\odot$ from equation (C.8) against the DE431-derived series used elsewhere in this paper, together with the residual error over one year.]

C.6 Remarks

- Equation (C.8) removes the last remaining ephemerides-dependent component of the DBSI model. Combined with equations (2.6) and (B.1),

the full DBSI series (2.2)–(2.4) can now be evaluated for any epoch, or for any other planet with known e , ε , $\tilde{\omega}$, P_a , without external astronomical data, directly supporting the paleoclimate and other-planet applications discussed in Section 5.

- Because the sign convention for $E_{\odot}$ varies across the solar radiation and celestial mechanics literature, equation (C.8) should be spot-checked against the existing DE431-based $E_{\odot}$ series at a small number of reference dates before being adopted as the primary source in equation (2.5). The derivation above defines $E_{\odot}$ so that it is subtracted from the mean hour angle in equation (2.5), consistent with the standard astronomical convention in which the equation of time is positive when the true (apparent) Sun leads the mean Sun (e.g. early November) and negative when it lags (e.g. mid-February).
- The e^3 term omitted from (C.2) is the limiting error source in Table C.2 and could be added directly, following the same series-inversion procedure used to obtain (A.5)–(A.6) and (C.2), if sub-tenth-second accuracy over multi-millennial timescales (where e evolves and higher-order terms matter more) is required.

D. Parametrization of solar declination $\delta_{\odot}$ and $\cos(z_{\odot})$

Local geographic character of DBSI, as defined in equation 2.1, is due to the appearance of $\cos(z_{\odot})$, which governs the daily TSI variability. From spherical trigonometry the zenith distance as function of solar declination $\delta_{\odot}$ and hour angle $t_{\odot}$ for specific location with latitude φ is (see for example [Green 1985]):

$$\cos(z_{\odot}) = \sin(\varphi) \sin(\delta_{\odot}) + \cos(\varphi) \cos(\delta_{\odot}) \cos(t_{\odot}). \quad (D.1)$$

Further, the following relation between the solar declination and solar longitude can be derived from spherical geometry (see [Green 1985]):

$$\sin(\delta_{\odot}) = \sin(\varepsilon) \sin(l_{\odot}), \quad (D.2)$$

where ε is obliquity of Earth's rotation axis. In order to express the $\cos(\delta_{\odot})$, which appears in equation (D.1), in terms of solar longitude we use the Taylor series expansion of $\sqrt{1 - \sin^2(\delta_{\odot})}$:

$$\begin{aligned} \cos(\delta_{\odot}) = & 1 - \frac{\sin^2(\varepsilon)}{2} \sin^2(l_{\odot}) - \frac{\sin^4(\varepsilon)}{8} \sin^4(l_{\odot}) - \frac{\sin^6(\varepsilon)}{16} \sin^6(l_{\odot}) \\ & - \frac{5 \sin^8(\varepsilon)}{128} \sin^8(l_{\odot}) - \frac{7 \sin^{10}(\varepsilon)}{256} \sin^{10}(l_{\odot}) - O((\sin(\varepsilon) \sin(l_{\odot}))^{12}) \dots \end{aligned} \quad (D.3)$$

Powers of $\sin(l_{\odot})$ can be expressed in terms of simple cosine functions with increasing multiples of the solar longitude yielding:

$$\begin{aligned} \cos(\delta_{\odot}) = & \Xi_0 + \Xi_2 \cos(2l_{\odot}) - \Xi_4 \cos(4l_{\odot}) + \\ & + \Xi_6 \cos(6l_{\odot}) - \Xi_8 \cos(8l_{\odot}) + \Xi_{10} \cos(10l_{\odot}) + \dots \end{aligned} \quad (D.4)$$

where:

$$\begin{aligned}
\Xi_0 &= 1 - \left[\frac{\sin^2(\varepsilon)}{4} + \frac{3 \sin^4(\varepsilon)}{64} + \frac{5 \sin^6(\varepsilon)}{256} + \frac{175 \sin^8(\varepsilon)}{16384} + \frac{441 \sin^{10}(\varepsilon)}{65536} + \dots \right] \\
\Xi_2 &= \frac{\sin^2(\varepsilon)}{4} + \frac{1 \sin^4(\varepsilon)}{16} + \frac{15 \sin^6(\varepsilon)}{512} + \frac{140 \sin^8(\varepsilon)}{8192} + \frac{1470 \sin^{10}(\varepsilon)}{131072} + \dots \\
\Xi_4 &= \frac{1 \sin^4(\varepsilon)}{64} + \frac{3 \sin^6(\varepsilon)}{256} + \frac{140 \sin^8(\varepsilon)}{16384} + \frac{420 \sin^{10}(\varepsilon)}{65536} + \dots \\
\Xi_6 &= \frac{\sin^6(\varepsilon)}{512} + \frac{20 \sin^8(\varepsilon)}{8192} + \frac{315 \sin^{10}(\varepsilon)}{131072} + \dots \\
\Xi_8 &= \frac{5 \sin^8(\varepsilon)}{16384} + \frac{35 \sin^{10}(\varepsilon)}{65536} + \dots \\
\Xi_{10} &= \frac{7 \sin^{10}(\varepsilon)}{131072} + \dots \\
&\dots
\end{aligned} \tag{D.5}$$

It is apparent that coefficients in (D.5) converge quickly as increasing powers of $\sin(\varepsilon)$ get progressively smaller than 1. Equations (D.2) and (D.4) are uniquely determining the solar declination as a function of obliquity and solar longitude. They are also viable alternative to expansion of the Sun's declination into Fourier series from [Spencer 1973] (see equation (D.6)), which is a popular choice by solar radiations modelers [Iqbal 1983].

$$\begin{aligned}
\delta_{\odot} &= 0.006918 - 0.39912 \cos(\Gamma) + 0.070257 \sin(\Gamma) - 0.006758 \cos(2\Gamma) \\
&\quad + 0.000907 \sin(2\Gamma) - 0.000907 \cos(3\Gamma) + 0.001480 \sin(3\Gamma) \\
\Gamma &= \frac{2\pi(d-1)}{365}, \quad d = 1 \dots 365
\end{aligned} \tag{D.6}$$

The accuracy of solar declination calculated by these approximations is estimated by comparing them with the solar declination downloaded from dynamical ephemerides DE431 [Folkner et.al. 2014], covering the period from January 01, 2024 to January 01, 2025. The solar longitude approximation (A.6) and obliquity (required in formulae (D.2) and (D.4)) is set to $\varepsilon = 23.^{\circ}44$.

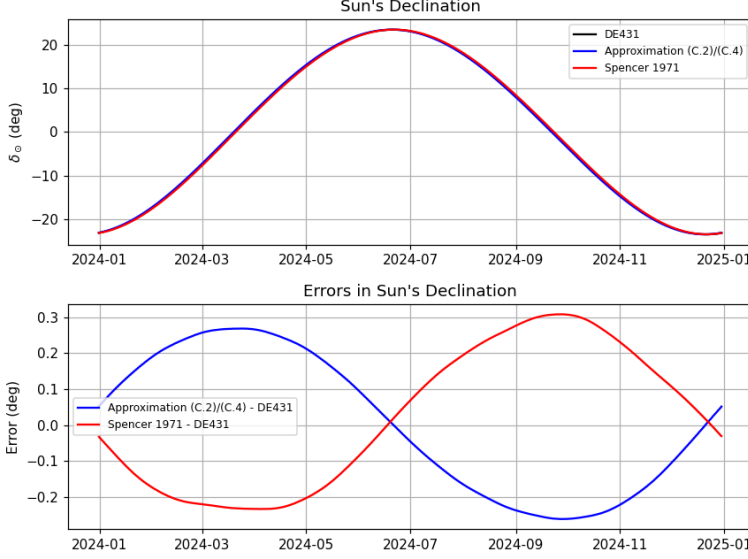


Figure 11: *Benchmarking of Different Approximations of Solar Declination*

Figure 11 (top) shows the the solar declination calculated by three different approaches discussed, while the same figure (bottom) shows the errors of the two methods relative to the most accurate DE431 based solar declination. It can be observed from Figure 11 that [Spencer 1973] formula error remains in range $[-0.^{\circ}23, 0.^{\circ}31]$ while declination by (D.2) and (D.4) varies in the range $[-0.^{\circ}26, 0.^{\circ}27]$ – the two approaches achieve comparable overall accuracy, of order a few tenths of a degree. This is a notable result: [Spencer 1973]’s formula is an empirical Fourier fit tuned directly against historical ephemeris data, whereas (D.2) and (D.4) are derived from first principles – Earth’s orbital elements and obliquity alone – with no such tuning against observed declination. Matching a dedicated empirical fit’s accuracy without any curve-fitting supports the physical validity of the orbital-element-based approach used throughout this paper.

Finally the cosine of the Sun’s zenith distance angle, parametrically dependent on latitude, obliquity, and orbital elements, is a function of time via solar longitude and local solar hour angle becomes:

$$\begin{aligned} \cos(z_{\odot}) = & \sin(\varphi) \sin(\varepsilon) \sin(l_{\odot}) + \\ & \cos(\varphi) [\Xi_0 + \Xi_2 \cos(2l_{\odot}) - \Xi_4 \cos(4l_{\odot}) + \Xi_6 \cos(6l_{\odot}) - \\ & \Xi_8 \cos(8l_{\odot}) + \Xi_{10} \cos(10l_{\odot}) + \dots] \cos(t_{\odot}) \end{aligned} \quad (D.7)$$

The solar hour angle is given by equation (2.5). In it, an equation of time parameter, which accounts for irregularities in Earth's rotation is present. The equation of time is a difference between the true and mean solar hour angle, depends on Earth's obliquity and orbital elements. It varies seasonally in range $[-15min+17min]$, and for the purpose of this study was downloaded from dynamical ephemerides DE431. Since orbital elements do not change significantly on short time scale, it is sufficient to select one annual curve and use it in determination of Sun's hour angle.

E. Parametrization of functions $S_{l_\odot}$ and $T_{l_\odot, t_\odot}$

In this section the DBSI (2.2) is expanded in simple, fast converging trigonometric series using results in Appendices A, B, C and D. Analytical expressions for frequencies, amplitudes and phases of the first four terms in trigonometric series (2.1) are calculated as functions of orbital elements and obliquity. After inserting expressions (B.1) and (D.6) into (2.2), the DBSI takes the following form:

$$DBSI_{t,\lambda} = L_{t,\lambda} \left(\frac{R_\odot}{a} \right)^2 \Theta_{\lambda,z_\odot} \times [E_0 + E_1 \cos(l_\odot - \tilde{\omega}) + E_2 \cos(2(l_\odot - \tilde{\omega}))] \times [\sin(\varphi) \sin(\varepsilon) \sin(l_\odot) + \cos(\varphi)(\Xi_0 + \Xi_2 \cos(2l_\odot) - \Xi_4 \cos(4l_\odot) + \dots) \cos(t_\odot)], \quad (E.1)$$

The equation (E.1) is an infinite trigonometric series, with the parameters $E_0 \approx 1$, $E_1 \propto e$ and $E_2 \propto e^2$ dependent on eccentricity only, and $\Xi_0 \approx 1 > \Xi_2 > \Xi_4 > \dots$ dependent on obliquity only. These properties of parameters in (E.1) can be utilized to create an 'ordered' trigonometric series of simple trigonometric functions in solar longitude and hour angle. In order to transform equation (E.1) into simple trigonometric form (2.2) all products of trigonometric functions of solar longitude and local hour angle should be expressed as sums and differences of trigonometric functions. After algebraic manipulation (E.1) takes the following form:

$$DBSI_{t,\lambda} = L_{t,\lambda} \left(\frac{R_\odot}{a} \right)^2 \Theta_{\lambda,z_\odot} \left[\sin(\varphi)[S_0 + S_1 \sin(l_\odot - l_1) + S_2 \sin(2l_\odot - \tilde{\omega}) + S_3 \sin(3l_\odot - 2\tilde{\omega})] + \cos(\varphi)[T_0 \cos(t_\odot) + T_1(\cos(t_\odot - l_\odot + \Phi_1) + \cos(t_\odot + l_\odot - \Phi_1)) + T_2(\cos(t_\odot - 2l_\odot + \Phi_2) + \cos(t_\odot + 2l_\odot - \Phi_2)) + T_3(\cos(t_\odot - 3l_\odot + \Phi_3) + \cos(t_\odot + 3l_\odot + \Phi_3)) + T_4(\cos(t_\odot + 4l_\odot - \Phi_4) + \cos(t_\odot - 4l_\odot + \Phi_4)) + \dots] \right] \quad (E.2)$$

After grouping the like-terms the amplitudes of functions $S_{l_\odot}$ and $T_{l_\odot, t_\odot}$ in equation (E.2) take the following form:

$$\begin{aligned}
S_0 &= -\frac{E_1 \sin(\varepsilon) \sin(\tilde{\omega})}{2} = -0.006478 \\
S_1 &= \sin(\varepsilon) \sqrt{E_0^2 + \frac{E_2^2}{4} - E_0 E_2 \cos(2\tilde{\omega})} = 0.398091 \\
S_2 &= -\frac{E_1 \sin(\varepsilon)}{2} = -0.0066524 \\
S_3 &= \frac{E_2 \sin(\varepsilon)}{2} = 0.000028 \quad (E.3)
\end{aligned}$$

and

$$\begin{aligned}
T_0 &= (E_0 \Xi_0 + \frac{E_2 \Xi_2}{2} \cos(2\tilde{\omega})) = 0.959850, \\
T_1 &= \frac{E_1}{2} \sqrt{\Xi_0^2 + \frac{\Xi_2^2}{4} + \Xi_0 \Xi_2 \cos(2\tilde{\omega})} = 0.015732 \\
T_2 &= \sqrt{\frac{E_0^2 \Xi_2^2}{4} + \frac{E_2^2 \Xi_0^2}{4} + \frac{E_2^2 \Xi_4^2}{16} + \frac{E_0 E_2 \Xi_2}{2} \left(\Xi_0 - \frac{\Xi_4}{2} \right) \cos(2\tilde{\omega}) - \frac{E_2^2}{4} \Xi_0 \Xi_4 \cos(4\tilde{\omega})} = 0.020580, \\
T_3 &= \frac{E_1}{4} \sqrt{\Xi_2^2 + \Xi_4^2 - 2\Xi_2 \Xi_4 \cos(2\tilde{\omega})} = 0.000348, \\
T_4 &= \frac{1}{2} \sqrt{E_0^2 \Xi_4^2 + E_2^2 \Xi_2^2 - E_0 E_2 \Xi_2 \Xi_4 \cos(2\tilde{\omega})} = 0.000223, \\
&\dots \quad (E.4)
\end{aligned}$$

while the phases of trigonometric functions appearing in (2.3)–(2.4) become:

$$\tan(l_1) = -\frac{(E_2/2) \sin(2\tilde{\omega})}{E_0 - (E_2/2) \cos(2\tilde{\omega})} \implies l_1 = 359^\circ.99823, \quad (E.5)$$

$$\begin{aligned}
\tan(\Phi_1) &= \tan(\tilde{\omega}) \frac{\Xi_0 - \Xi_2/2}{\Xi_0 + \Xi_2/2} \implies \Phi_1 = 103.^{\circ}7179, \\
\tan(\Phi_2) &= \frac{(E_2/2)(\Xi_0 + \Xi_4/2) \sin(2\tilde{\omega})}{E_0\Xi_2/2 + E_2/2(\Xi_0 - \Xi_4/2) \cos(2\tilde{\omega})} \implies \Phi_2 = 359.^{\circ}9172, \\
\tan(\Phi_3) &= \tan(\tilde{\omega}) \frac{-\Xi_2 + \Xi_4}{\Xi_2 + \Xi_4} \implies \Phi_3 = 77.^{\circ}1097, \\
\tan(\Phi_4) &= \frac{(E_2\Xi_4/4) \sin(2\tilde{\omega})}{E_0\Xi_4/2 + E_2\Xi_4/4 \cos(2\tilde{\omega})} \implies \Phi_4 = 180.^{\circ}164, \\
&\dots \\
&\text{(E.6)}
\end{aligned}$$

Values of amplitudes and phases (E.3)–(E.6) are obtained from mean values in year 2022 of the DE431 published eccentricity $e = 0.0167141$ and perihelion longitude $\tilde{\omega} = 103.^{\circ}1609$.

Frequencies of terms in (E.2) with the largest amplitudes are daily (term T_0) and annual (term S_1), while all other periodic terms contain frequencies which are linear combinations of daily and annual frequencies:

$$t_{\odot} \pm nl_{\odot} = f_n t \mp n\Delta\Phi + L - E_{\odot}, n = 0, 1, 2, \dots \quad (\text{E.7})$$

where frequencies are:

$$f_n = \frac{2\pi}{P_d} (1 \pm n \frac{P_d}{P_a}), n = 0, 1, 2, \dots \quad (\text{E.8})$$

with $P_d/P_a \approx 1/365 \approx 0.002739$. $\Delta\Phi$ term is

$$\Delta\Phi = \tilde{\omega} + \frac{2\pi}{P_a} t_p + 2e \sin\left(\frac{2\pi}{P_a} (t - t_p)\right). \quad (\text{E.9})$$

F. Atmospheric Transmissivity Function $\Theta_{t,\lambda,z_\odot}$

So far in this paper, there was no discussion on transmissivity function $\Theta_{\lambda,z_\odot}$. Even though the focus of paper is on orbital and rotational aspects of the solar irradiance modeling, some form of transmittance function is required for benchmarking calculations. Equation (F.1) shows optical depth model of the spherical-shell, homogeneous atmosphere with vertically stratified density detailed derivation and analysis of this model will be presented in a separate paper. In this model $R_\oplus$ is the Earth's radius, H_a is the atmospheric thickness, ϱ_0 is density at the surface, and h_T is altitude of a point on surface at which transmissivity is calculated. Further, κ_λ and σ_λ are absorption and scattering cross sections in the beam, respectively. In this simple model absorption and scattering cross sections are assumed independent on altitude and optical path. For the purposes of calculation in this paper, the atmospheric thickness is assumed to be 100 km, surface density 1.22 kg/m³, while the extinction cross section (absorption plus scattering out of beam) is set to 0.00250 m²/kg – tuned against a pooled Golden+Izaña set of ground-based clear-sky measurements, as described in Appendix F.1 below (Section 4 reports the resulting validation).

$$\tau_\lambda(z_\odot, h_T) = \varrho_0(\kappa_\lambda + \sigma_\lambda) \int_0^{h_T} \frac{e^{\frac{-(H_a-x)}{H_a}} dx}{\sqrt{1 - \left(\frac{R_\oplus}{R_\oplus + H_a - x} \sin(z_\odot) \right)^2}} \quad (F.1)$$

The integral in equation (F.1) can be solved numerically to get the optical depth as a function of altitude and Sun's zenith distance, which leads to the expression for transmissivity in spherical-shell model of the atmosphere.

$$\theta_\lambda(z_\odot, h_T) = e^{-\tau_\lambda(z_\odot, h_T)} \quad (F.2)$$

Figure 12 below shows two transmissivity functions, one for the plane-parallel atmosphere and one for the spherical-shell atmosphere (obtained from (F.1)). Figure 13 shows mild sensitivity of transmissivity functions to the altitude, in the $z_\odot \in [0^\circ, 60^\circ]$ range.

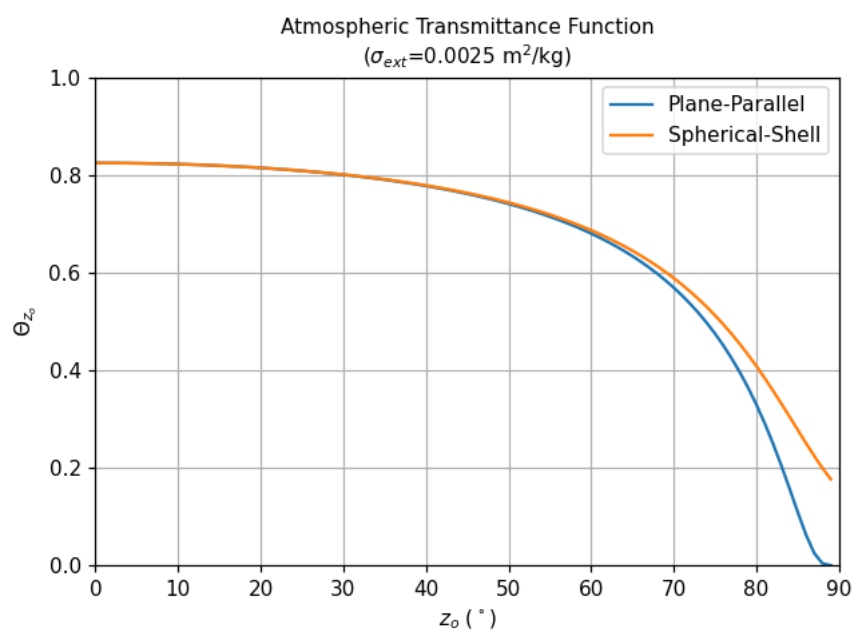


Figure 12: *Simple Transmissivity of Spherical-Shell and Plane-Parallel Model Atmospheres.*

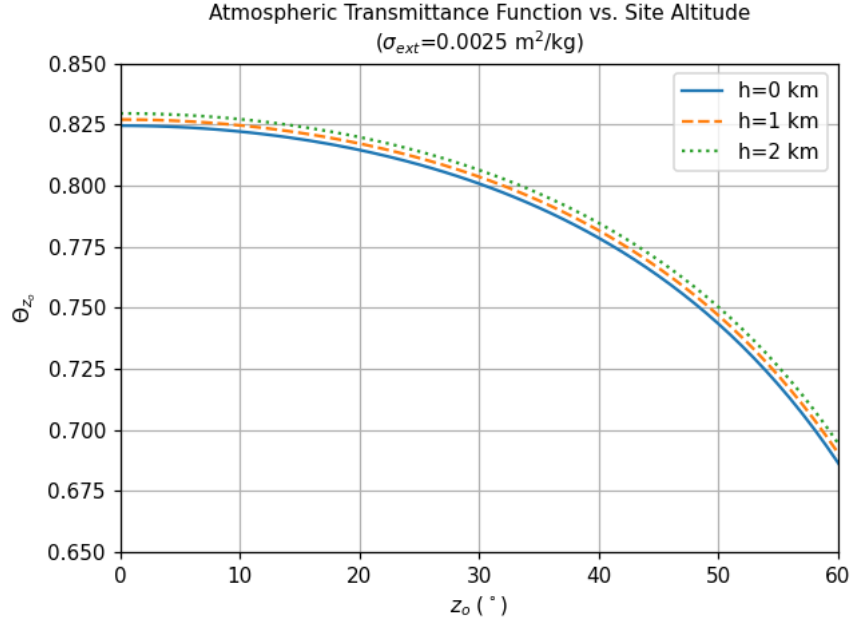


Figure 13: *Atmospheric Transmissivity at Different Altitudes.*

F.1 Tuning σ_{ext} , H_a Against Ground-Based Measurements

Equation (F.1) has three free constants: atmosphere thickness H_a , surface density ρ_0 , and extinction cross section σ_{ext} . Of the three, only ρ_0 has an independent physical referent (the ICAO/US Standard Atmosphere sea-level value, 1.225 kg/m^3) and was held fixed at 1.22 kg/m^3 ; H_a and σ_{ext} are both unsourced stand-ins (the former nominally associated with the Kármán line, $\sim 100 \text{ km}$) and were investigated as tuning parameters against the ground-based clear-sky validation (Section 4).

σ_{ext} was originally tuned (with H_a held at 100 km) against Golden, CO alone, giving a clean, well-behaved interior minimum at $\sigma_{\text{ext}} = 0.00295 \text{ m}^2/\text{kg}$ (RMSE = 22.2 W/m^2 , down from 39.1 W/m^2 at the previous default, $0.0035 \text{ m}^2/\text{kg}$). Once Izaña was added as a third validation site (Section 4), that single-site choice was revisited: Golden's own optimum turned out not to generalize – evaluated at Izaña, $\sigma_{\text{ext}} = 0.00295$ gives RMSE = 42.6 W/m^2 , well above Izaña's own standalone optimum, $\sigma_{\text{ext}} = 0.00200 \text{ m}^2/\text{kg}$ (RMSE = 11.7 W/m^2). Since Golden and Izaña are the two sites whose targets are

direct-beam-only (Alderville’s G1 includes diffuse irradiance the model does not produce, biasing a single-site fit against it – Section 4), σ_{ext} is instead tuned here against a *pooled* Golden+Izaña objective: a single RMSE computed over the concatenated residuals of both sites together. Figure 14 shows RMSE and $|\text{bias}|$ against σ_{ext} at all three sites individually (Alderville shown for reference, not included in the pooled objective); the pooled Golden+Izaña minimum falls at $\sigma_{\text{ext}} = 0.00250 \text{ m}^2/\text{kg}$ (pooled RMSE = 27.0 W/m^2 , versus 37.2 W/m^2 if Golden’s own 0.00295 optimum were kept). This is the value adopted in equation (F.1) above and used throughout Section 4’s Table 3; it is a deliberate compromise, not an improvement for either site individually – at $\sigma_{\text{ext}} = 0.00250$, Golden’s own RMSE worsens to 36.0 W/m^2 while Izaña’s improves to 21.3 W/m^2 (Table 3). Alderville’s own standalone optimum, $\sigma_{\text{ext}} \approx 0.00200 \text{ m}^2/\text{kg}$, is broadly consistent with Izaña’s, though not itself pooled into the objective for the reason above.

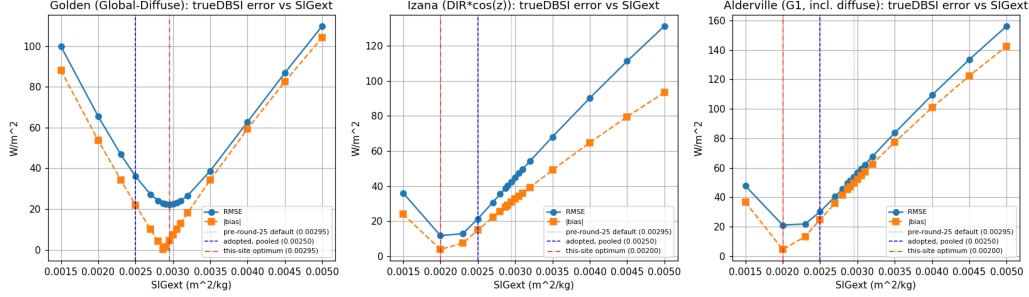


Figure 14: σ_{ext} sensitivity at all three validation sites: RMSE and $|\text{bias}|$ of trueDBSI against each site’s own clear-sky target, as a function of σ_{ext} (H_a , ϱ_0 held fixed). Gray dotted line: value adopted here ($0.00250 \text{ m}^2/\text{kg}$, pooled Golden+Izaña optimum); red dash-dot line: each site’s own individual optimum. Golden’s minimum is broad and shallow; Izaña’s and Alderville’s are both sharp, near $0.00200 \text{ m}^2/\text{kg}$.

A joint sweep allowing both H_a and σ_{ext} to vary together was also carried out (against Golden only, predating the Izaña extension above), to check whether a second free parameter meaningfully improves on the single-parameter result. Figure 15 shows the resulting RMSE surface: a diagonal valley of near-equally-good fits running from short- H_a /high- σ_{ext} to long- H_a /low- σ_{ext} , rather than a localized minimum – the same kind of near-degeneracy already evident analytically between ϱ_0 and σ_{ext} , which enter the optical depth (F.1)

as a bare product and so cannot be separated by a single-site fit. A genuine (if broad and shallow) interior optimum along that valley does exist, near $H_a \approx 300$ km, $\sigma_{\text{ext}} \approx 0.0008\text{--}0.0010$ m²/kg (RMSE = 15.9 W/m²) – but it was *not* adopted: reaching it requires H_a to move $3\times$ past the value’s own stated physical rationale, for an RMSE improvement ($22.2 \rightarrow 15.9$ W/m²) that is real but modest set against how far outside the parameter’s intended meaning it requires going. This is offered as a diagnostic finding in its own right: a second free “shift” parameter, added to a model whose knobs (Appendix F, main text) can only rescale the transmittance curve, not change its shape, buys only marginal RMSE at the cost of physical interpretability – it does not, and structurally cannot, resolve the shape mismatch discussed next in Appendix F.2. H_a was therefore left at its original 100 km value, with only σ_{ext} tuned (subsequently against the pooled Golden+Izaña objective above, superseding this figure’s single-site square marker).

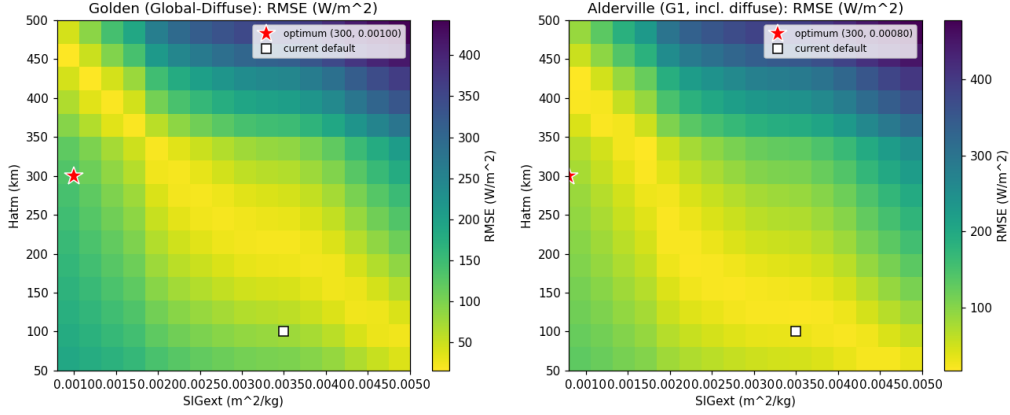


Figure 15: Joint H_a – σ_{ext} RMSE surface against Golden, CO (left) and Alderville, ON (right), showing the diagonal near-degenerate valley described in Appendix F.1. Star: joint optimum; square: value adopted in equation (F.1) ($H_a = 100$ km held fixed, σ_{ext} tuned).

F.2 Comparison with the Bird & Hulstrom (1981) Model

The transmittance model above collapses the atmosphere’s whole spectral response into one broadband, altitude-independent extinction cross section – an explicit simplification, per the discussion at the start of this Appendix. To gauge what a structurally richer, but still broadband (not spectral), al-

ternative buys, the direct-beam transmittance of Bird & Hulstrom (1981) [Bird & Hulstrom 1981] was implemented and compared against the same validation sites (Section 4). Rather than one lumped exponential, it factors Θ into five independently-shaped processes – Rayleigh scattering, ozone absorption, uniformly-mixed-gas absorption, water-vapor absorption, and aerosol extinction – each a closed-form function of Kasten’s empirical relative air mass and, for aerosol and water-vapor absorption, of measured/typical site inputs (turbidity, precipitable water, ozone column, surface pressure) rather than a single fitted constant.

Neither validation dataset records the Bird & Hulstrom model’s own required inputs (surface pressure, precipitable water, ozone column, aerosol turbidity) for its clear-sky day. Surface pressure was set from the site’s own altitude via the ISA barometric formula; ozone column (0.35 atm-cm) and precipitable water (1.4 cm) were left at typical mid-latitude literature values (not tuned). Aerosol turbidity – flagged by [Bird & Hulstrom 1981] itself as the single largest source of day-to-day uncertainty in the model – was tuned the same way as σ_{ext} in Appendix F.1: a single scalar scale factor applied to both spectral turbidity inputs together (their ratio held fixed, since a single clear-sky day cannot constrain the aerosol’s spectral shape independently), swept against Golden. Unlike the H_a – σ_{ext} surface above, this sweep has a clean interior minimum: turbidity at $0.5\,\mu\text{m}$ tuned from a generic 0.10 down to 0.030 (Figure 16) reduces Golden’s RMSE from 41.9 to 7.1 W/m² – notably below this paper’s own tuned model (Table 3, now pooled against Golden+Izaña rather than Golden alone – see Appendix F.1), even though only one Bird & Hulstrom parameter was adjusted, matching the one adjusted here. The fitted turbidity, while low, sits below (not outside) the 0.10–0.51 range [Bird & Riordan 1984] measured across several Golden clear-sky days, consistent with 2024-03-09 having been an unusually clean day rather than the fit being physically unreasonable. (Alderville’s own apparent optimum, zero aerosol, was not used – an artifact of its target including diffuse irradiance neither model produces, not a genuine atmospheric finding.)

A natural question, given that σ_{ext} above is now tuned against a pooled Golden+Izaña objective (Appendix F.1), is whether the same pooling should be applied to Bird & Hulstrom’s turbidity. It should not, and the reason

is itself informative: unlike σ_{ext} , which has a genuine interior optimum at both sites individually, Izaña’s own RMSE for Bird & Hulstrom’s aerosol term decreases *monotonically* as turbidity is reduced to zero – the same boundary-optimum behavior already noted above for Alderville, but now at the one site whose target is otherwise direct-beam-only and free of the diffuse-irradiance contamination flagged there. A pooled Golden+Izaña objective is accordingly dominated by Izaña’s much larger absolute errors and collapses to $\tau_{a,0.5} = 0$, which is not a genuine compromise the way the pooled σ_{ext} is: Golden’s own RMSE at that point rises from 7.1 to 31.4 W/m² – pooling here would sacrifice Golden’s well-behaved interior optimum to chase Izaña’s boundary value, rather than average two comparable optima. Notably, even at zero aerosol Izaña’s fit retains a substantial negative bias (−22.4 W/m², RMSE 37.1 W/m²), which a supplementary check attributes to precipitable water rather than aerosol: with turbidity held at zero, lowering precipitable water from the generic mid-latitude default used here (1.4 cm) toward ~ 0.2 –0.3 cm – physically plausible for a dry, high-altitude site – removes most of the residual bias (RMSE 37 $\rightarrow$ 13 W/m²). This suggests Izaña’s poor fit under Bird & Hulstrom is better explained by its un-tuned, generic precipitable-water input than by aerosol turbidity, and that a fair pooled comparison to σ_{ext} would require jointly tuning more than one Bird & Hulstrom parameter per site – beyond the scope of the single-parameter turbidity sweep performed here. Turbidity is accordingly left tuned against Golden alone, as originally reported, rather than pooled.

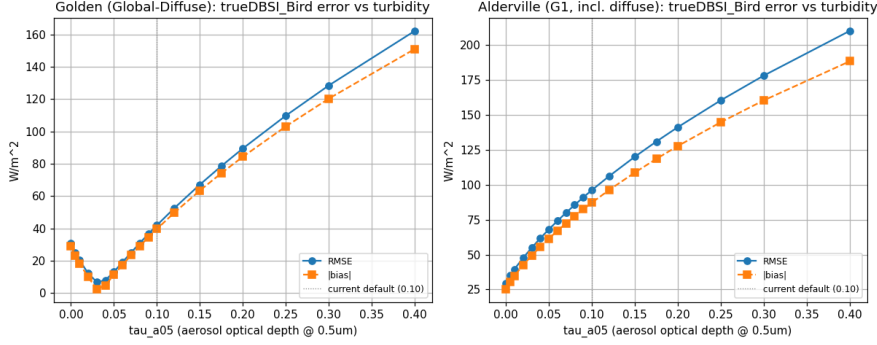


Figure 16: *Bird & Hulstrom (1981) RMSE and $|bias|$ vs. aerosol turbidity at $0.5\mu\text{m}$, Golden (left) and Alderville (right). Golden’s clean interior minimum, at $\tau_{a,0.5} = 0.030$, is the value used in Table 3 and Figure 5.*

The more informative result is not the aggregate RMSE but the residual’s shape (Figure 5, Section 4): at Golden, this paper’s own Θ – now tuned to a pooled Golden+Izaña objective rather than to Golden alone (Appendix F.1) – leaves a systematic, airmass-dependent residual that is large and positive through the middle of the day (+55 to +67 W/m² near solar noon) and falls toward small, slightly negative values near sunrise, reflecting that the pooled $\sigma_{\text{ext}} = 0.00250\text{ m}^2/\text{kg}$ under-attenuates relative to Golden’s own single-site optimum. Appendix F.1 shows no re-tuning of H_a , ϱ_0 or σ_{ext} , singly or jointly, changes this residual’s basic *shape* (only its overall level) – because all three only ever rescale a single Beer-Lambert exponential, never bend its shape. The Bird & Hulstrom model, still fit to Golden alone, remains markedly flatter and smaller across the day. A plausible mechanism, noted by [Bird & Riordan 1984] in the course of developing a fully spectral model: real absorption bands (water vapor, uniformly mixed gases) follow a *saturating* functional form rather than a pure exponential – attenuation per unit path length falls off as the path lengthens (the “curve of growth” in radiative transfer) – which a single constant-cross-section exponential cannot reproduce at any parameter value. This is offered as a well-evidenced candidate mechanism, not a proof, and – together with the tuning results of Appendix F.1 – is left for the companion paper [Ćatović & Žaknić-Ćatović 2026] on radiation transport through the atmosphere, along with a fully spectral treatment, rather than expanded further here: the objective of the present paper is the

parametric $(1/r_{\odot})^2 \cos(z_{\odot})$ expansion of Section 2, and $\Theta_{t,\lambda,z_{\odot}}$ is retained here as a deliberately simple, clearly-flagged placeholder (Table 3, Section 4, quantifies how much of the ground-comparison residual is attributable to that placeholder, as opposed to the geometric expansion itself).